

Binary to Gray Code converter report

Badsha

September 5, 2026

Contents

1	What is Binary to Gray and How to convert Binary to Gray ?	3
1.1	For a 4bit converter:	3
2	Binary to Gray Conversion Rule	3
2.1	For a 4-bit binary numbers:	3
2.2	The gray code is generated using XOR Operation:	3
3	Implementation	3
3.1	Thus, the circuit requirements:	3
3.2	Verilog Implementation:	4
3.3	Compact Implementation:	5
4	Example	5
5	Verification	5
5.1	Example:	5
6	Used Tools	5

1 What is Binary to Gray and How to convert Binary to Gray ?

An binary-to-gray Code converter is a combinational Logic circuit that convert a binary number into its equivalent gray code.

1.1 For a 4bit converter:

- Input: 4-bit binary numbers
- Output: 4-bit Gray code

2 Binary to Gray Conversion Rule

2.1 For a 4-bit binary numbers:

Binary: B3 B2 B1 B0 Gray: G3 G2 G1 G0

2.2 The gray code is generated using XOR Operation:

```
1      G3 = B3 ;
2      G2 = B3 ^ B2 ;
3      G1 = B2 ^ B1 ;
4      G0 = B1 ^ B0 ;
```

Therefore,

```
1      G = B ^ (binary >> 1);
```

3 Implementation

3.1 Thus, the circuit requirements:

- 3x XOR gates
- 1x Direct Connection

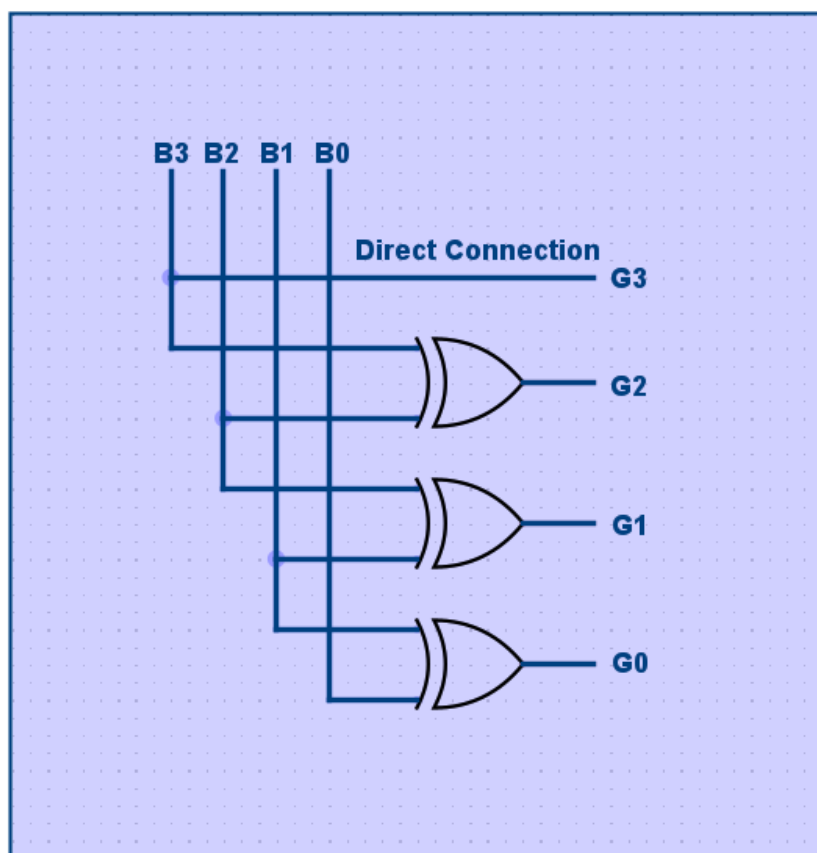


Figure 1: Binary to Gray Logic Diagram

3.2 Verilog Implementation:

```

1      module BinarytoGray(
2          input  [3:0] B,
3          output [3:0] G
4      );
5
6          assign G[3] = B[3];
7          assign G[2] = B[3] ^ B[2];
8          assign G[1] = B[2] ^ B[1];
9          assign G[0] = B[1] ^ B[0];
10
11      endmodule

```

3.3 Compact Implementation:

The same circuit can also be written using the Binary to gray equation:

```
1      content...module BinarytoGray(  
2          input wire [3:0] B,  
3          output wire [3:0] G  
4      );  
5  
6          assign G = B ^ (B >> 1);  
7  
8      endmodule
```

Both implementation produce same result.

4 Example

Binary = 1010

```
1          G3 = B3 (1) Direct Connection 1 = 1  
2          G2 = B3 ^ B2 1 ^ 0 = (1)  
3          G1 = B2 ^ B1 0 ^ 1 = (1)  
4          G0 = B1 ^ B0 1 ^ 0 = (1)
```

Therefore, Binary = 1010 Gray = 1111

5 Verification

The design can be verified using verilog testbench. The testbench applies different 4bit binary inputs and checks whether the generated gray code is correct.

5.1 Example:

6 Used Tools

- Verilog hdl
- Icarus verilog

Binary	Gray	Decimal
0000	0000	0
0001	0001	1
0010	0011	2
0011	0010	3
0100	0110	4
0101	0111	5
0110	0101	6
0111	0100	7
1000	1100	8
1001	1101	9
1010	1111	10

Table 1: The complete 4-bit truth table can be used to verified all 16 possible input combinations

- GTKWave
- VS code
- Linux/WSL

Made by [@DESOS](#)